The Field Grade Leader
Organizational Leadership in the US Army

Terrain Model Construction – A Guest Post by David Chichetti

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Terrain Model construction is an often undervalued step to enhance shared understanding of the mission. For a combined arms rehearsal (CAR), a good terrain model is necessary to enhance collaboration and dialogue requisite for good planning and unity of effort.[1] But both the literature and training on this skill are thin. The March 1998 CALL Newsletter "Rehearsals" has good information, but could be updated to meet the requirements of the modern battlespace. Captains Career Courses and the Command and Staff College have not dedicated curriculum to this subject. At the Brigade, Division and Corps levels, site construction is often last in planning priorities. This results in the purchase of large-scale maps or simply arraying plotter pictures of objectives. These techniques are expedient but do not accurately convey the challenges of terrain. As a result, rehearsals can suffer, sometimes causing confusion or even embarrassment. To better convey the commander's intent, units must build an intricately detailed terrain model to provide clear visualization.

Most leaders lack experience in large terrain model construction and pass this task to junior officers or NCOs. But when the rehearsal site needs to show the entire area of influence combat enablers, and specific micro details, middle management, needs to be involved. The intent of this article is to provide some overarching guidelines and lessons learned for terrain model construction. It is intended to supplement current literature and to share lessons learned, specifically focusing on-site construction process. Preparation, foundation construction, terrain building, and presentation create the framework for this discussion. Techniques shared in this article will strike the balance between time available for construction and the complexity of construction.[2]

Overarching Considerations

The objective, to put in colloquial terms, is to create the "HGTV" effect, it has to "pop." Leaders should be able to discern the battlespace at first glance; the terrain model must be easy on the eyes and intuitive. As stated before, the terrain model needs accurate detail, specifically 3-dimensional detail. It should demonstrate elevation, hydration, and feature key graphic control measures. It should encompass the entire area of influence.[3] Lastly, it must be functional. Briefers and icon movers will need to be able to work around and through the site. Unit leaders will want to walk the terrain so it cannot be too fragile. A terrain model package should be expeditionary so that it can be easily unpacked and set up to fit any terrain. The more modular your terrain model package is, the easier it will be to set up. These broad considerations are a good baseline to guide the thought process. The next step is to do some initial preparations.

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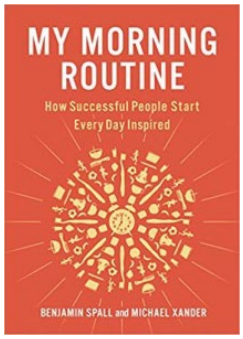

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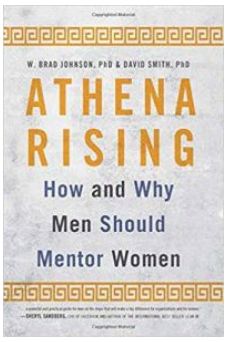
The Preparation Phase

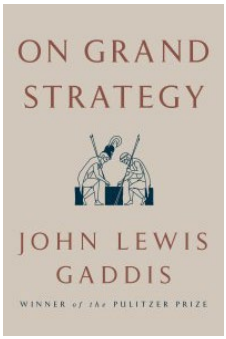
Good preparation can save time, resources, and ultimately a good deal of frustration. The preparation phase is the best opportunity to get leadership involvement. Ensure that the map you are using is the same one that the commander will use during the actual rehearsal. The BDE or DIV Geospatial Intelligence (GEOINT) team can assist with this task. After securing a map, ask your leadership and or planners good questions. Leadership will designate the key pieces that must be visible to the audience. A few questions to ask are: what terrain am I modeling? What is the area of influence? What is key terrain? Which routes and graphic control measures are required? The goal of the preparation phase is to ensure CAR site is not at odds with your commander's visualization of the battlefield. Once you have this initial guidance, you can build your plan and timeline for construction.

So, what materials should you use? There are dozens of different types of materials that can be utilized for construction. I have seen Christmas lights, sound effects and even Soldiers in blue T-shirts "fight" Soldiers in red T-shirts. The point is that you should use whatever it takes to achieve shared understanding. The following are some factors to consider. Materials should be malleable, rugged and modular so that they can be manipulated to form terrain features or graphic controls. For our terrain model, my team used multiple different colors of 100 mph tape, green spray paint, chicken wire, burlap, 20' X 30' brown tarps, 20' X 30' blue tarps and numerous rolls of engineer tape. The tarps and engineer tape are important for scaling. Using terrain you are operating in is a good technique but tarps can be made to look like any terrain, survive weather hazards and will save time by knowing your precise measurements. Now that materials and leadership involvement has been secured, the next phase is building a foundation.

Building the Foundation

Building the 2-dimensional foundation is the starting point. Disclaimer: I am not an artist (if you couldn't tell already), but my wife has a Master's degree in Fine Arts. As she watched me racking my brain over this issue, she provided some useful artist tricks to help me build the model, particularly in regards to scaling geographic features. If you use tarps as your 2-dimensional base, you already know the size and dimensions. For example, my team knew our terrain model would always be either a square 60' x 60' model or 40' x 60' rectangle depending on the direction of travel for blue forces. This can easily be replicated if terrain is the base. Once you know the measurements of the geographic area, you can begin to apply a temporary grid for scaling. A good technique for this is to adhere white engineer tape on the model creating a grid (figure b).





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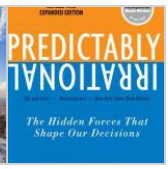

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Figure B: Temporary grid lines for scaling

Building temporary grid squares allows you to create a two-dimensional outline of the map to scale. The foundation should not focus on terrain, but instead roads, phase lines, kill boxes, basically all permanent graphic control measures. Ensure you create an essential pathway or pathways to allow for presenters to move icons in dense terrain (mountains, cities). The foundation provides the layout of the scheme of maneuver from a 2-dimensional perspective. The next step is developing 3-dimensional features.

Terrain Building

3–dimensional terrain building is difficult but is important for accurate visualization. The complexity and time involved in making 3-dimensional terrain is usually the root cause of why people choose to push the easy button and buy a large map. These maps can be ordered through your local training support office for about $5,000 and restrict you to a certain operational environment. But it is difficult to demonstrate the complexity of multiple domains with only a 2-dimensional map. For example, a Joint Forcible Entry Operation (JFE) needs to show terrain considerations to illuminate potential issues with time, distance and line of sight. A wet gap crossing needs to show the size of consolidation areas, hydration and overwatch locations. For airspace, being able to demonstrate mean sea level (MSL) versus above ground level (AGL) can be critical to managing the shaping fight. Creating 3-dimensional terrain gives command teams the depth and width of the battlespace. It is essential that units take time to develop this technique.

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Terrain features should match the operational environment. Whichever technique is used, the site construction team should keep in mind that personnel may need to walk on it or place icons on it. Terrain features should not be taller than mid-shin level height so that the audience can view the entirety of the operational environment (OE) without visual obstruction. To build mountains, our team used chicken wire, burlap and garden containment liner Chicken wire can be easily manipulated to form any terrain. When densely packed, it is resilient enough that if stepped on, it will move back to its original shape. Once we had modeled the chicken wire, we added garden liner or burlap to cover it. The next step was to “ghillie” the terrain. Vegetation (tree branches, pine needles, etc.) or the crinkle paper purchased from craft stores worked well to represent wooded terrain. Similarly, shredded paper can work just as easily. (figure c).

Figure C: Burlap and containment sheet over chicken wire with shredded paper for vegetation

For key hydration areas, the crinkle/shredded paper works well because it differentiates the river or lake from the rest of the environment. Finally, spray paint can be used to not only camouflage inconsistencies but color terrain accurately. If you are not using tarps as your 2-dimensional foundation, a few Soldiers with E-tools can easily forage the forest for dirt, pine needles or any other vegetation required. For urban areas, MRE boxes and materials can be used to represent key buildings. Construction teams should consider windows, doors and other apertures especially for company and below missions. Site construction teams should add as much detail as possible to ensure the terrain matches the actual environment. Ultimately, 3-dimensional terrain detail assists the audience with understanding the broad strokes of the operation.

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Presentation

The presentation piece of the terrain model process deals with how the audience views the battlespace. This is not merely using different colors, but shapes, sizes, and numbers for both friendly and enemy forces. Signs that represent objectives, urban areas, phase lines should all have different symbols to allow the audience to discern graphic control measures and terrain. This is also a good point in the terrain building process to bring in the rest of the staff for improvements. For example, my team would lay down all terrain then the fires team would overlay PAAs, kill boxes etc. The following are some techniques for enhanced audience visualization:

(a.) Writing phase lines vertically and objectives horizontally.

Figure D: Different shapes for different graphic control measures

(b.) A method to show the timeline and task organization displayed to provide context for the audience.

(c.) To show airspace coordination, place a plotter sheet or tarp in the background of the terrain model and list out each air control measure and max ordinates (figure e).

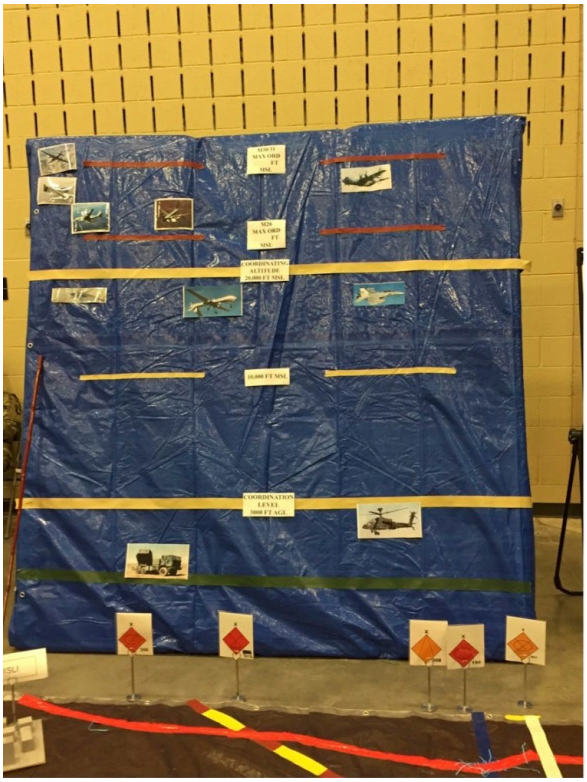

Figure E: Airspace coordination board

(d.) A “bleeding” icon technique for enemy and friendly icons is a way to show attrition of forces (figure f).

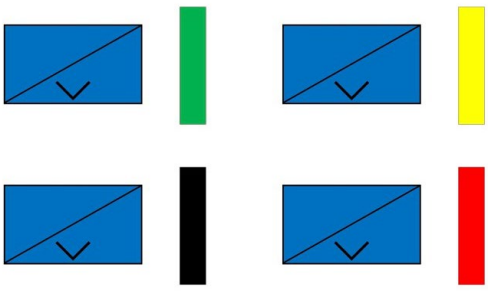

Figure F: An example “bleeding” icon template

(e.) A projector screen that follows the script of the rehearsal can enhance situational understanding.

These techniques are just some examples of how you can ensure common understanding amongst the team.

Conclusion

Terrain model construction is a challenging process that requires leadership involvement for quality control. Whether you are in an expeditionary environment or in a controlled environment in the consolidation area, every effort should be made to guarantee shared understanding. Essentially, do whatever it takes to get the point across. As a field grade leader, your commanders are relying on you to ensure the staff collectively gives input to site construction. The above process of preparation, building the foundation, adding terrain features, and focusing on presentation is a simple framework to focus your efforts. Good plans, well-rehearsed can create a common understanding. A visually dynamic and accurate terrain model can enhance your execution.

Major David Chichetti is an infantry officer, a graduate of The Citadel and holds a master's degree in Security Studies from Kansas State University. He served as an XO at the battalion and brigade levels for 2BCT, 101st ABN (AASLT). Maj Chichetti is currently serving is as the 101st ABN (AASLT) Chief of Current Operations.

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References:

[1] ADP 6-0 Mission Command

[2] "Rehearsals." *Center for Army Lessons Learned Newsletter* 98, no. 5 (March 1998): 22-23.

[3] ibid

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Please note: I reserve the right to delete comments that are offensive or off-topic.

2 thoughts on "Terrain Model Construction – A Guest Post by David Chichetti"

Mark M says:
March 7, 2018 at 1:55 am

Terrain models are great and definitely improve the planning and rehearsal process for any organization. These are some great suggestions for building a terrain model.

In the compressed planning timelines found at the CTCs, though, terrain models of this size and detail may be a stretch for most BCT and below organizations. What is the timeline for building a terrain model of these size shown about (40×60)?

Mike Pappal, Dir CALL says:
March 7, 2018 at 9:29 pm

The March 1998 CALL Newsletter "Rehearsals" is in the final stages of update. Expect it to be available in the very near future.

Comments are closed.

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